

Probabilistic databases

A database is an organised collection of data. Database technology represents a fundamental tool for storing and accessing data in applications ranging from managing customer and employee data in enterprises to managing increasingly larger amounts of measurement readings in scientific experiments. Life without databases seems unimaginable these days. How else could one book flights or hotels, or do business in a large city like London?

Whilst they are clearly useful and ubiquitous, today's database management systems require data to adhere to a rigid and predefined structure and to be complete and trustworthy. Great effort is invested in cleaning enterprise data, a labor-intensive and complex process that accounts for 30 per cent (80 per cent of the development time in a data warehouse project). It is not surprising that this costs US businesses billions of dollars annually.

At the same time, internet applications witness an unprecedented shift towards unstructured and user-generated content. These data come in vast quantities, originate from diverse sources without a common structure, and are often inaccurate and contradictory. Although they do not fit in the rigid frame of traditional databases, these data are valuable. In particular, marrying traditional databases that host accurate and carefully maintained data with the more dynamic and up-to-date Web data is desirable. Consider, for instance, business intelligence (BI), whose goal is to extract and analyse business data by mining a large collection of databases. BI systems can be made

more useful by including external data such as twitter feeds, blogs, or email messages, in order to extract valuable business information. By analysing blogs or twitter feeds and joining them with offline databases of products, companies can obtain early feedback about the quality of a new product, or its degree of adoption; such knowledge is very valuable, both for manufacturers and for investors.

Probabilistic Data

Making sense of unstructured and uncertain Web data is not an easy task. Using information extraction techniques, one can give a probabilistic meaning to such data. This is the case for the Never-Ending Language Learning (NELL) system, which extracts triples (Subject, Predicate, Object) from Web data and assigns confidence scores to learned facts. For instance, NELL concludes from some source with 100% confidence that Angelina Jolie, is, actor and Angelina Jolie, is, female; however, from a different source it derives with 96.9% confidence that Jolie is actually male (as of May 2011). The reasons for this ambiguity are straightforward: Information extraction from natural language sources is error-prone, and different data sources can convey contradictory information.

The Google Squared service takes this idea one step further: It aggregates web data into uncertain tables that describe a given keyword. The table for the keyword "Bollywood movies" is depicted in the figure. For the movie "Sholay", it proposes Hindi and English as possible lan-



Item Name	Language	Director	Release Date	Genre
Hindi Language for Sholay	Hindi	Ramprasad	15 August 1975	Comedy
English Language for Sholay	English	Ramprasad	15 August 1975	Comedy
Hindi Language for Sholay	Hindi	Ramprasad	15 August 1975	Comedy
English Language for Sholay	English	Ramprasad	15 August 1975	Comedy
Hindi Language for Sholay	Hindi	Ramprasad	15 August 1975	Comedy
English Language for Sholay	English	Ramprasad	15 August 1975	Comedy

guages, and Ramesh Sippy and R. D. Burman as possible directors. Alternative values are annotated with confidence scores: Google's ranking suggests that Sippy is more likely the director than Burman, and that Hindi is more likely the language than English.

Both NELL and Google Squared extract uncertain tables from unstructured web data, which is a first key step to making the data truly accessible to structured query languages like SQL. In both systems, the uncertainty in the data is represented by confidence scores. Using probabilistic database management systems, one can join standard or uncertain tables and retrieve the answers ranked according to their likelihood. In our example, we can join the Bollywood movies square with a square Hollywood movies in order to retrieve a list of directors including Shekhar Kapur who have produced both Bollywood and Hollywood movies; or we can find currently popular movies by joining our square with live twitter trends.

Very recent years have witnessed an impressive body of research on managing probabilistic data, early prototype systems, and applications in information extraction, RFID and scientific data management, and financial risk assessment. This is described at length in the first book on this topic. 



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Robert Fink is a doctoral student and Dan Olteanu is a University Lecturer, both in the Department of Computer Science at the University of Oxford. Dan's group works on theoretical and system aspects of probabilistic databases, and recently included three bright Indian computer scientists: Smitha Mysore-Shankar, Swaroop Rath and Yuvika Suvarna.